Xi'an Jiaotong-Liverpool University
西交利物浦大学

CPT202

Software Engineering Group Project
Department of Computing Stage 3 | Level 2

SECTION A: Basic Information

Brief Introduction to the Module

Students will work in groups of four members to produce a working software system. The deliverables and working methods will be prescribed. To provide experience of group working; To provide experience of all aspects of the development of a moderately sized software system; To prepare students for their individual projects in the final year; To consolidate material from the first semester of the third year.

Key Module Information

Module name	Module code	Credit value	Semester in which the module is taught	Pre-requisites needed for the module
Software Engineering Group Project	CPT202	5	SEM2	N/A
Programmes on which the module is shared	BSc Information and Computing Science BSc Information Management and Information Systems			

Module Leader and Contact Details

Module Leader:

Name	Email address	Office telephone number	Room number	Office hours	Preferred means of contact
CHENGTAO JI	Chengtao.Ji@xjtlu.edu.cn	89165344	SD467(SIP Campus-Science Building)	Wednesday 10:00 AM - 11:00 AM, Thursday 10:00 AM - 11:00 AM	Email
Brief Biography	My research mainly focuses on data mining, information visualization, and machine learning, which are applied to various scenarios, e.g., astronomy, medicine, digital humanity, etc.				

Additional Teaching Staff and Contact Details:

Role	Name	Email address	Office telephone number	Room number	Office hours	Preferred means of contact
co-lecturer	Soon Phei Tin	Soon.Tin@xjtlu.edu.cn	81889038	SD531(SIP Campus-Science Building)	Thursday 09:00 am – 11:00 noon	Email
co-lecturer	Nanlin Jin	Nanlin.Jin@xjtlu.edu.cn	88161800	SD429(SIP Campus-Science Building)	Tuesdays 15pm-16pm; Thursday 11am-12noon	Email

SECTION B: What You Can Expect from the Module

Educational Aims of the Module

Students will work in groups of four to produce a working software system.
The deliverables and working methods will be prescribed.
to provide experience of group working;
to provide experience of all aspects of the development of a moderately sized software system;
to prepare students for their individual projects in the final year;
to consolidate material from the first semester of the third year.

Learning Outcomes

Students completing the module should be able to:

A. Work as part of a development team demonstrating effective communication and interpersonal skills to design and develop a software system.
B. Demonstrate an understanding of the software development process including the principal methods and issues involved in deploying systems t o meet business goals.
C. Specify the requirements of a software system.
D. Understand the role of properly written documentation in the process of software development.
E. Recognize the legal, social, ethical and professional issues involved in the development and deployment of a software system.

Methods of Learning and Teaching

The overall strategy is to allow self and peer guided learning within a tightly defined framework.

At the beginning of the module an introductory lecture will outline details of the project scheme, and documentation detailing the framework and e xpectations will be provided. Thereafter lectures will be given at the rate of one a week describing the key skills needed to carry out the project. Students will be put into teams of four, and thereafter will be expected to work largely autonomously. Teams will be expected to hold regular proj ect meetings, the minutes of which will be monitored by staff. Certain software deliverables will be prescribed: the projects will produce a databas e front end tools for maintaining the database tools for accessing, analysing and presenting the data

The domain of the system must be selected by the team. Each project will have three milestones:requirements specification, design specification, c ompleted system and final report

Each of these will be reviewed and assessed by staff. Staff will be available on an 'as needed' basis to offer support, guidance and to arbitrate any difficulties.

Syllabus & Teaching Plan

Week Number	Mode of Delivery(Lecture/Tutorial/Seminar/Field Trip/Other)	Topic	Pre-reading and others

W1	Lecture	• Module requirements
		• Project setup
		• Scrum Framework
		• Release of videos:
		  ◦ Scrum Product Backlog, Requirements, and User Stories, Sprint, Sprint Planning, Sprint Execution
		  ◦ Azure Project, Azure Boards
			Complete the forming of groups.

W2	Lecture	Scrum Framework	Complete the allocation of the project.
W3	Lecture	• Workshop: Intro to Spring Boot framework. • Workshop: Database Programming	

W4	Lecture	• Workshop: UI template for Webapp • Workshop: Spring Security • Source code management with Git Repository	Assignment 1 Due
W5	Lecture	Report writing and Documentation	
W6	Lecture	Software Testing	
W7	Lecture	Consultation Week	
W8	Lecture	Application Deployment	
W9	Lecture	Workshop: Legal, social, ethical, and professional issues	
W10	Lecture	Project Sharing	Assignment 2 Due
W11	Lecture	Project Sharing	
W12	Lecture	Project Sharing	
W13	Lecture	Project Sharing	Assignment 3 Due

Assessment Details

Initial Assessment

Assignment 1-User Requirement Specification (20% of the module mark)

Assessment Type: *CW*
Learning outcomes assessed: *A,B,C,D*
Duration: *N/A*
Resit opportunity: *N*

Assessment Task	Learning Outcomes	Weighting	Release Date	Due Date
Assignment 1-User requirement specification	A,B,C,D	20%	24/Feb/2025	16/Mar/2025
Generative AI Permissions	The use of Generative AI for content generation is not permitted on this coursework.			
Requirement and Guideline of the Assessment Task				

Objective:

To collaboratively analyze project requirements and independently develop well-defined Product Backlog Items (PBIs) using the Scrum framework.

Coursework Process:

1. **Group Discussion and Requirement Analysis:**
   - Form groups to discuss and understand the project requirements thoroughly.
   - Collaboratively identify and outline the key modules of the project.
2. **Module Assignment:**
   - Within your group, assign each module to a different group member.

    - Ensure each member clearly understands their assigned module and its requirements.
3. **Independent PBI Development:**
    - Each student will independently develop PBIs for their assigned module.
    - Use the standard PBI format, including user stories and acceptance criteria, to ensure clarity and testability.
4. **Peer Review Process:**
    - Upon completion of PBI development, participate in a peer review process.
    - Evaluate a peer's PBIs, providing constructive feedback focused on clarity, precision, and alignment with Scrum principles.
5. **Final Report Submission:**
    - Compile a final report that includes the following sections:

Final Report Components:

1. **Understanding of the Scrum framework**
    - Provide a clear and concise explanation of the Scrum framework, emphasizing the role of PBIs in the process.
    - Strictly limit this section to 100 words.
2. **Pre-Groomed PBI**
    - Develop five pre-groomed PBIs relevant to the module assigned to you, each with a user story and acceptance criteria.
    - Ensure each PBI is well-defined and adheres to the PBI format.
3. **Collaboration and Reflection**
    - **Peer Review and Feedback:**
        - Document your participation in the peer review process.
        - Provide constructive feedback to peers, focusing on clarity, precision, and alignment with Scrum principles.
        - Feedback should be actionable, helping peers improve their work.
        - Evaluation criteria include constructiveness, depth of analysis, and clarity of communication.
        - Strictly limit this section to 250 words.
    - **Reflection on Learning:**
        - Detailed reflection on what you learned from the coursework and the peer review process.
        - Discuss insights gained, personal growth, and how feedback was incorporated.
        - Strictly limit this section to 100 words.

Submission Guidelines:

- Submit your final report as a single PDF document via Learning Mall by the above submission date, including all sections outlined in the Final Report Components above.
- Ensure your PBIs are clearly formatted and easy to read.
- DO NOT submit more than 5 PBIs. Marks will only be given to the first 5 PBIs in your report.
- Include your peer review feedback and personal reflection as separate sections in the document.
- Name the report as CPT202-A1-StudentID.pdf

Marking Scheme

1. **Understanding of Scrum and PBI Format (10 Marks)**
    - **Explanation of Scrum Framework (5 Marks)**:
        - Clear and concise explanation of the Scrum framework, focusing on the role of PBIs in the development process.
    - **Correct Use of PBI Format (5 Marks)**:
        - Adherence to the standard PBI format, including user stories and acceptance criteria.
2. **Quality of PBIs (20 Marks)**
    - **Clarity and Precision (10 Marks)**:
        - PBIs are clearly written and easily understandable by all stakeholders.
    - **Logical and Relevance (10 Marks)**:
        - PBIs are logically relevant to the assigned module.
3. **Pre-Groomed PBI (30 Marks)**

- **PBI 1 (6 Marks)**:
  - Well-defined user story and acceptance criteria.
- **PBI 2 (6 Marks)**:
  - Well-defined user story and acceptance criteria.
- **PBI 3 (6 Marks)**:
  - Well-defined user story and acceptance criteria.
- **PBI 4 (6 Marks)**:
  - Well-defined user story and acceptance criteria.
- **PBI 5 (6 Marks)**:
  - Well-defined user story and acceptance criteria.

4. **Collaboration and Reflection (40 Marks)**
   - **Peer Review and Feedback (25 Marks)**:
     - Students are required to participate in a peer review process, providing constructive feedback on another student's PBIs.
     - Marks are awarded based on the quality, thoughtfulness, and specificity of the feedback provided.
     - **Criteria for Evaluation**:
       - **Constructiveness (10 Marks)**: Feedback should be constructive, offering actionable suggestions for improvement.
       - **Depth of Analysis (10 Marks)**: Feedback should demonstrate a deep understanding of the PBIs and Scrum principles.
       - **Clarity and Communication (5 Marks)**: Feedback should be clearly communicated and well-structured.
   - **Reflection on Learning (15 Marks)**:
     - Students must submit a detailed reflection (Approximately 100 words) on what they learned from the assignment and the peer review process.
     - Marks are awarded based on the depth of insight, personal learning, and application of Scrum principles articulated in the reflection.
     - **Criteria for Evaluation**:
       - **Insightfulness (5 Marks)**: Reflection should provide deep insights into the learning process and understanding of Scrum.
       - **Personal Growth (5 Marks)**: Reflection should highlight personal growth and how feedback was incorporated.
       - **Feedback and Application (5 Marks)**: Reflection should demonstrate how feedback from peer review were applied or understood through the coursework.

Assignment 2-Group Report (30% of the module mark)

Assessment Type: *CW*
Learning outcomes assessed: *A,B,D,E*
Duration: *N/A*
Resit opportunity: *N*

Assessment Task	**Learning Outcomes**	**Weighting**	**Release Date**	**Due Date**
Assignment 2-Group report	A,B,D,E	30%	24/Feb/2025	27/Apr/2025
Generative AI Permissions	The use of Generative AI for content generation is not permitted on this coursework.			
Requirement and Guideline of the Assessment Task				

General instruction

By week 10, your team should have already implemented 3 software increments. Therefore, you are required to make the first release of the software. It is expected that you have already completed the major system architecture, system-level software design, some main features, and various testing for 3 increments.

In this assignment, your team is required to write a group report and deliver a presentation.

- The group report must be in the format given in the Group Report Template, which is provided in the Assignment Package. It is important to note that the report must be at most 12 pages, excluding appendixes.
- The presentation is expected to be recorded in a 10-minute video. In the presentation, provide the summary of the work conducted to arrive at the first release of the software. The software used in the software demonstration must be accessed from the URL pre-allocated to y

our team. The URL must be made clear at the beginning of the software demonstration. For verification purposes, the software from the URL must remain unchanged for 1 week after the submission of the presentation.

The marking criteria and the weighting is based on the following factors: -

A. Group Report

1. Report quality (15%):

   The report should be well presented. The language used in the report must be clear and concise. Whenever appropriate, use images or drawings to aid understanding.

2. Introduction (15%):

   The introduction presented in the report must sufficiently show your

   understanding of the project background and initial requirements.

   The project background should be described from general to specific. It must include a problem statement, project aims, user characteristics, project scope, assumptions and dependencies, and project risks that are clearly outlined.

   A use case diagram should be presented to help describe the project scope. It must be written within the scope of the requirements at the same time to avoid obvious omissions.

3. Architectural Design (25%):

   The overview of the architectural design should be clearly stated and justified.

   The system architecture details should be provided, and an overall architectural diagram should be presented to represent the system's structure.

   An ERD of the structure of your database should be provided with a sufficient explanation of the structure.

4. Software Design (25%):

   Show a high-level process flow for each essential function, illustrating how the software is modularized. It is better to add diagrams and code fragments to clearly describe these services.

   Describe the supporting services your software may have used, i.e., Git, Docker, maven, etc., and explain how these services help you during the development.

   Explain the coding standards, structure, and conventions followed in your project, ensuring they are logical, easy to understand, easy to follow, and efficient.

   Describe the production environment clearly.

5. Software Testing (10%):

   How the software testing is conducted should be described in terms of unit testing, integration testing, and acceptance testing. The testing process, testing data, and results (screenshots) also should be included. Make sure to show sufficient test data and test results.

6. Conclusion and Future work (10%):

   A conclusion should be given regarding what your group has done, what your group has achieved, and what problems are in your software for now. Future work should be given as well in terms of what work your group will do and related timeline if possible.

B. Presentation

1. Understanding of the project (40%):

The presentation should show a good understanding of the project in terms of the project background, architectural design, software design, and database.

2. Software Demonstration (40%):

    The software demonstration should cover all the aspects of the software, including front-end and back-end processes, as well as database access. The demonstration should proceed in a use case with close-to-real data and emphasize the core use case.

3. Presentation Skills (20%):

    The presentation should demonstrate effective communication at an appropriate pace. Any slides used should have a clear layout and proper use of visual effects. Figures are well presented, and the presentation is well timed.

Individual contribution

We recognize that each member of the team may contribute differently to the group report and the presentation. Therefore, you are required to fill out, sign, and submit an Individual Contribution Form, which is provided as part of the Assignment Package. The contribution (%) in the form indicates the overall contribution of an individual on this assignment. The team leader should arrange a meeting to discuss and complete the form in the presence of all members. The contribution percentage for everyone must be agreed upon by all other members. In the case where a contribution cannot come to an agreement, the team could seek help from the module leader. Please note that only works that can be assigned with the due date can be considered as contributions to the assignment, for example: -

- Writing the report or presentation slide
- Presenting or demonstrating in the presentation
- Editing the report or video

The following works are not to be considered contributions: -

- Participating in the discussion
- Providing ideas and comments.
- Providing financial support.

Submission and markings notes:

- Your team is expected to submit the following documents: -
    - Group report (PDF format; 12 pages max with a font size 10)
    - Presentation slide (PDF format)
    - Presentation video (10 minutes max, 100MB max)
    - Source code in a **<u>single</u>** Zip file
    - Individual contribution form (PDF format)
- Name your documents accordingly: -
    - Group report: CPT202-A2-Report-***GroupXXX***.pdf
    - Presentation slide: CPT202-A2-Slide-***GroupXXX***.pdf
    - Presentation video: CPT202-A2-Video-***GroupXXX***.mp4
    - Source code: CPT202-A2-Code-***GroupXXX***.zip
    - Individual contribution form: CPT202-A2-Contribution-***GroupXXX***.pdf
- Each team must submit ONLY one copy of the above documents by the team leader via Learning Mall
- Group Report 20% + Presentation 10% = 30%
- When preparing your source code, take note that: -
    - Files in proprietary or non-standard file formats may not be readable and so are likely to be ignored.
    - Do not submit the compiled binary, which will cause your zip file to be overly large.

Assignment 3-Individual Report (50% of the module mark)

Assessment Type: *CW*
Learning outcomes assessed: *A,B,D,E*
Duration: *N/A*
Resit opportunity: *N*

Assessment Task	**Learning Outcomes**	**Weighting**	**Release Date**	**Due Date**
Assignment 3-Individual report	A,B,D,E	50%	24/Feb/2025	18/May/2025
Generative AI Permissions	The use of Generative AI for content generation is not permitted on this coursework.			
Requirement and Guideline of the Assessment Task				

General instruction

In this assignment, you are required to write an individual report to describe *your contribution* to the final solution for the group project. This report should only include your own work, NOT the group work.

The report must be in the format given in Individual Report Template, which is provided in the Assignment Package. It is important to note that the report must be <u>at most</u> 8 pages, excluding the following: (1) the cover page that has the student ID, (2) the page Table of Content, (3) references, and (4) appendixes. The appendixes may include examples of PBIs, UML, Entity-relationship diagram (ERD), screenshots of the products, and/or (part of) source code, etc.

The software solution that you have designed and implemented must be accessible from the URL pre-allocated to your team. The URL must be made clear on the cover page of the report. For verification purposes, the software from the URL must remain unchanged for 4 weeks after the submission of the presentation.

The marking criteria and the weighting are based on the following factors: -

1. Report quality (15%):

   The report should be well presented, ensuring that the language used is clear, concise, and free from ambiguity. Utilize visual aids such as images, diagrams, and charts to enhance comprehension where appropriate. Ensure that all visual elements are properly labelled and referenced within the text.

2. Software development process (20%):

   The understanding of how Scrum facilitates the software development process and effective communication should be well discussed. The discussion should revolve around the parts you did (e.g., use a specific example from your own parts).

3. Software Design (25%):

   The general design and implementation of all the PBIs that you have individually developed should be presented generally. Provide a clear justification for these designs and implementations. Use UML diagrams to support your illustrations.

   A detailed discussion on 2 essential PBI(s) that you have contributed should be provided in terms of the following aspects:

   a. Logical flow: you could present and discuss an activity/sequence diagram.

   b. Database: discuss how the database design supports the PBI.

   c. States of the objects involved. Discuss how the state changes and affects the objects' behaviors.

   d. UI. Discuss how the UI design helps deliver the PBI.

4. Change Management (15%):

   Discuss how the requirements changes should be handled in a Scrum Project. The discussion should revolve around the project you do (e.g., use a specific example from your own project). If there are no requirements changed in your project, you should consider this part from the perspective of the whole project or potential changes in future work.

5. Legal, social, ethical, and professional issues (15%)

   Discuss where you have considered or the project has been influenced by legal, social, ethical, and professional issues. Provide the possi ble solutions to these issues. The discussion should revolve around the project you do (e.g., use a specific example from your own projec t). If the parts you did do not involve such issues, you should consider this part from the view of the whole project.

6. Conclusion (10%)

   Clearly illustrate the lessons you have learned from this project. Also, provide the improvements that can be made in the future from your perspective.

Submission and markings notes:

- You must submit the report in PDF format via Learning Mall by the above submission date. Name the report as CPT202-A3-StudentID.p df.
- Strictly keep the report to a maximum of 8 pages with a font size of 10, excluding appendixes.

Resit Assessment

There is no resit assessment for this module

Reading Materials

Type	Title	Author	ISBN/Publisher
Mandatory Textbo oks	N/A		
Optional Textbook s	N/A		
Reference Textbo oks	PROJECTS IN COMPUTING AND IN FORMATION SYSTEMS	C. W. DAWSON	9781292073460/PRENTICE HALL
	PATTERNS OF ENTERPRISE APPLIC ATION ARCHITECTURE	MARTIN FOWLER	9780321127426/ADDISON WESLE
	ESSENTIAL SCRUM: A PRACTICAL GUIDE TO THE MOST POPULAR AG ILE PROCESS	KENNETH RUBIN	9780137043293/ADDISON WESLE
Additional Materia ls			

SECTION C: Additional Information

This section provides students with essential information and resources pertaining to their academic studies to ensure a successful academic journey and engage ment with the module.

Student Feedback:

The University is committed to receiving and responding to student feedback in order to improve the quality of the student experience within the in stitution. It is University policy that the preferred way of doing this is by using the Online Student Module Feedback Questionnaire Survey. Studen ts are encouraged to complete the questionnaire survey for this module at the end of the semester.

Attendance:

The University expects students to attend all timetabled learning sessions associated with this module, and to engage with the relevant learning and support resources. Student attendance will be recorded using the Attendance Management System (AMS). Please follow your teacher’s instructi ons for recording your attendance at each session. Students are responsible for managing their attendance, and should take prompt action to infor m the Module Leader in case circumstances beyond their control affect their class attendance. You are advised to read the University's 'Student Attendance Policy' for more information.

Rules of Submission for Assessed Coursework:

The University has detailed rules and procedures governing the submission of assessed coursework. You need to be familiar with the rules and procedures as detailed in the University's 'Code of Practice on Assessment'.

Late Submission of Assessed Coursework:

The University attaches penalties to the late submission of assessed coursework. You need to be familiar with the rules as detailed in the University's 'Code of Practice on Assessment'.

Mitigating Circumstances:

Students who experience serious illness or other unforeseen circumstances as defined in the Mitigating Circumstances (MC) Policy that prevent them from submitting coursework or taking final/resit exams on time can apply for coursework deadline extension or final/resit exam authorized absence. The application should be made before the assessment date under Academic Records page on e-Bridge. Misuse of the MC policy will result in disciplinary actions and demerit points.

Academic Integrity:

The University aims to foster a learning environment which produces students who embrace academic integrity, understand that they must produce their own work, are able to acknowledge explicitly any material that has been included from other sources or legitimate collaboration, and to present their own findings, conclusions or data based on appropriate and ethical practice. Any violation of academic integrity including plagiarism, collusion, copying, submission of commissioned or procured work, and/or falsification and fabrication of data will result in penalties and demerit points. Please be familiar with the University's Academic Integrity Policy.

Examination Misconduct:

The University also values academic integrity in the conduct of examinations. Any behavior that violates examination regulations will not be tolerated and will result in penalties and demerit points, as detailed in the policy of Regulations for Conduct of Examinations.

Student Discipline Point System:

Any violation of Academic Integrity Policy, Regulation for Conduct of Examinations, and abuse of the Mitigating Circumstances Policy will accrue demerit points. These points will be placed in the university system, and on the official XJTLU transcript. For details, please refer to the Student Discipline Point System appended to the Regulations for the Conduct of Examinations.

Generative AI:

Information on whether the use of Generative AI is permitted or not for each assessed coursework is indicated in the Assessment Details section of this module handbook.

For more information and resources on Generative AI and your learning and assessment, please consult the 'XJTLU AI for Learning' pages of the Learning Mall Core.

Learning Mall Core:

Copies of lecture notes and other materials are available electronically through the Learning Mall Core, the University's virtual learning environment, at learningmall@xjtlu.edu.cn.

Communication:

All official communication concerning module-related matters will be conducted via e-mail and/or as Learning Mall Core announcements. Other modes of electronic communication are treated as informal.

Further Support:

You are advised to contact your Module Leader in the first instance if you experience any issues with your learning on this module. You may also contact your Academic Advisor or Programme Director. Further information on the kinds of support that the University provides to students can be found in the XJTLU Student Handbook.

You are strongly advised to read the policies mentioned above very carefully, because this will help you perform better in your academic studies. You can find all the policies and regulations related to your academic study on the e-Bridge → 'Document Zone' page.

Generated from e-Bridge

Cut-off Date: 13/Feb/2025